

Calculus of Variations

Ahmed Alotaibi

Motivation

Assume we have an expression in the form

$$F(y, y', x)$$

where $y = y(x)$.

Now consider this integral

$$J := \int_a^b F(y, y', x) dx$$

for different $y(x)$ in the **function space**, we will get different value of the integral J .
Natural question is : for what $y(x)$ we get a minimum (or maximum) value for J ?

Function Spaces

We will consider each function $y(x)$ as a point in some **function space** .

Definition

The function space $\mathcal{C}^k(a, b)$ is a complete normed linear space (*Banach space*) consisting of all functions $y(x)$ defined on $[a, b]$ which have a continuous k th derivative.

- The norm in $\mathcal{C}^k(a, b)$ space is defined as

$$||y||_k = \sum_{i=0}^k \max_{a \leq x \leq b} |y^{(i)}|$$

- This allows us to define a continuous function whose domain is a function space.

Definition

By a **functional** we mean a correspondence which assigns a real number to each function in a function space . For $k = 1$

$$J[y] : \mathcal{C}^1(a, b) \longrightarrow \mathbb{R}$$
$$y(x) \longmapsto \int_a^b F(y(x), y'(x), x) dx$$

- An example of functionals is :
- the arc length

$$J[y] = \int_a^b \sqrt{1 + (y'(x))^2} dx$$

Euler-Lagrange equation

Theorem

consider the functional

$$J[y] = \int_a^b F(y(x), y'(x), x) dx$$

a necessary condition for $J[y]$ to have an extremum for a given function $y(x)$ is that $y(x)$ satisfy

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} = 0$$

*which is known as **Euler-Lagrange equation** .*

Example :shortest distance between two points

- Consider two points $A = (0, a)$ and $B = (1, b)$ find the shortest distance between them .
- we will search for a function $y(x)$ with $y(0) = a$ and $y(1) = b$
- Such that the arc length of $y(x)$ between the two points is minimum i.e. we need to minimize $J[y] = \int_0^1 \sqrt{1 + (y'(x))^2} dx$
- Apply *EL equation* we get $\frac{d}{dx} \left(\frac{y'}{\sqrt{1 + (y'(x))^2}} \right) = 0 \implies y' = k$ for some $k \in \mathbb{R}$
- This means $y(x) = kx + c$ by applying the boundary condition we get $y(x) = (b - a)x + a$, as expected !

The Variation of a Functional

- To prove the previous theorem, we first need to study some important concepts.
- One of such concepts is the **variation of a functional** ($\delta J[h]$) which plays a similar role as the differential of a real-valued function (df)

Definition

Let $J[y]$ be a functional defined on some function space, consider the increment

$$\Delta J[h] := J[y + h] - J[y]$$

where y is fixed. If we can write $\Delta J[h]$ as

$$\Delta J[h] = \phi[h] + \epsilon ||h||$$

where $\phi[h]$ is a **linear functional** and $\epsilon \rightarrow 0$ as $||h|| \rightarrow 0$. Then $J[y]$ said to be **differentiable** and we define the **variation** of $J[y]$ as

$$\delta J[h] := \phi[h]$$

Extremum Criterion for Functionals

Theorem

A necessary condition for the differentiable functional $J[y]$ to have an extremum for $y = \hat{y}$ is that its variation vanish for $y = \hat{y}$, i.e., that

$$\delta J[h] = 0$$

Proof. Suppose $J[y]$ has a minimum for $y = \hat{y}$. According to the definition of the variation $\delta J[h]$, we have

$$\Delta J[h] = \delta J[h] + \varepsilon \|h\|,$$

where $\varepsilon \rightarrow 0$ as $\|h\| \rightarrow 0$. Thus, for sufficiently small $\|h\|$, the sign of $\Delta J[h]$ will be the same as the sign of $\delta J[h]$. Now, suppose that $\delta J[h_0] \neq 0$ for some h_0 . Then for any $\alpha > 0$, we have

$$\delta J[-\alpha h_0] = -\delta J[\alpha h_0].$$

Hence, can be made to have either sign for arbitrarily small $\|h\|$. But this is impossible, since by hypothesis $J[y]$ has a minimum for $y = \hat{y}$, i.e.,

$$\Delta J[h] = J[\hat{y} + h] - J[\hat{y}] \geq 0 \quad \square$$

Fundamental Lemma

Lemma

Let $\varphi : [a, b] \rightarrow \mathbb{R}$ be continuous and such that

$$\int_a^b \varphi(t)\psi(t) dt = 0$$

for all functions $\psi \in C^1([a, b])$ satisfying $\psi(a) = \psi(b) = 0$. Then $\varphi \equiv 0$ on $[a, b]$.

Proof. Suppose not, that is, suppose that for some $t_0 \in (a, b)$ we have $\varphi(t_0) \neq 0$, say $\varphi(t_0) > 0$ (a similar argument works if $\varphi(t_0) < 0$). Since φ is continuous there is a small open interval $I = (t_1, t_2) \subset [a, b]$ about t_0 where $\varphi(t) \geq \frac{1}{2}\varphi(t_0) > 0$ for all $t \in I$. Choose a continuous ψ so that $\psi(t) = 0$ for t outside I , but with $\psi(t) > 0$ on a neighborhood of t_0 (for example, $\psi(t) = (t - t_1)(t_2 - t)$ for $t \in I$ and $\psi(t) = 0$ otherwise). Then

$$\int_a^b \varphi(t)\psi(t) dt = \int_I \varphi(t)\psi(t) dt \geq \frac{1}{2}\varphi(t_0) \int_I \psi(t) dt > 0,$$

which contradicts the assumption. Therefore, $\varphi \equiv 0$ on $[a, b]$.

The proof of Euler-Lagrange equation

We aim to find a function $y(x)$ that makes the functional

$$J[y] = \int_a^b F(x, y, y') dx$$

have an extremum, where the function satisfies the boundary conditions:

$$y(a) = A, \quad y(b) = B$$

Let us consider a small change $h(x)$ such that $h(a) = h(b) = 0$, and define:

$$y(x) \rightarrow y(x) + h(x)$$

Then the change in the functional is:

$$\Delta J = J[y + h] - J[y] = \int_a^b [F(x, y + h, y' + h') - F(x, y, y')] dx$$

Using Taylor expansion:

$$\Delta J = \int_a^b (F_y h + F_{y'} h' + \cdots) dx$$

The first-order (linear) part the variation δJ :

$$\delta J = \int_a^b (F_y h + F_{y'} h') dx$$

Using integration by parts on the second term and noting $h(a) = h(b) = 0$, we get:

$$\delta J = \int_a^b \left(F_y - \frac{d}{dx} F_{y'} \right) h dx$$

For this to be zero for all admissible h , we must have:

$$\frac{d}{dx} F_{y'} = F_y$$

